

MA 113 Mathematics 2

Lecture 1

Functions of More than One Variable

Limits and continuity

Definition 2: (Limit of a function)

If the function f in two variables x, y defined inside and on a circle with center (a, b) with the possibility of non – occurrence at (a, b) itself then the limit of $f(x, y)$ when (x, y) approaches (a, b) is L , and we write

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

Rule: If the limit of f at (a, b) differs at two different paths then the limit of $f(x, y)$ as $(x, y) \rightarrow (a, b)$ is not exist.

Example:

Determine if the following limits exist or not. If they do exist give the value of the limit.

$$(i) \lim_{(x,y,z) \rightarrow (2,1,-1)} 3x^3z + yz \cos(\pi x - \pi z)$$

$$(ii) \lim_{(x,y) \rightarrow (5,1)} \frac{xy}{x+y}$$

Solution:

$$\begin{aligned}(i) \quad \lim_{(x,y,z) \rightarrow (2,1,-1)} 3x^3z + yz \cos(\pi x - \pi z) \\ = 3(2)^2(-1) + (1)(2)\cos(2\pi + \pi) = -14\end{aligned}$$

$$(ii) \quad \lim_{(x,y) \rightarrow (5,1)} \frac{xy}{x+y} = \frac{(5)(1)}{5+1} = \frac{5}{6}$$

Theorem

The successive limits of the function $f(x,y)$ at the point (x,y) as it approaches (a,b) defined as:

$$\lim_{y \rightarrow b} \lim_{x \rightarrow a} f(x, y) = \lim_{x \rightarrow a} \lim_{y \rightarrow b} f(x, y)$$

If the successive limits of the two functions are not equal or one of them does not exist, then $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ does not exist.

Example

Using the method of repeated limits, discuss the existence of

the $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x-y)}{x+y}$.

Solution

Take the limit on the path $y = 0$, we get

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} \frac{\sin(x-y)}{x+y} \right) = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) = 1$$

Take the limit on the path $x = 0$, we get

$$\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} \frac{\sin(x - y)}{x + y} \right) = \lim_{y \rightarrow 0} \left(\frac{\sin(-y)}{y} \right) = \lim_{y \rightarrow 0} \left(-\frac{\sin(y)}{y} \right) = -1$$

Since the two limits are not equal, then $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ does not exist.

Example:

Using the method of repeated limits, discuss the existence of

the $\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2 + 4y^2 - 6xy}{2x^2 + y^2}.$

Solution:

Take the limit on the path $y = 0$, we get

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} \frac{2x^2 - 4y^2 - 6xy}{2x^2 + y^2} \right) = \lim_{x \rightarrow 0} \left(\frac{2x^2 - 0 - 0}{2x^2 + 0} \right) = \lim_{x \rightarrow 0} (1) = 1$$

Take the limit on the path $x = 0$, we get

$$\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} \frac{2x^2 - 4y^2 - 6xy}{2x^2 + y^2} \right) = \lim_{y \rightarrow 0} \left(\frac{0 - 4y^2 - 0}{0 + y^2} \right) = \lim_{y \rightarrow 0} (-4) = -4$$

Since the two limits are not equal, then $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ does not exist.

Limit along the Path:

Example:

Discuss the existence of the $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{x^2 - y^2}$.

Solution

Let $y = mx$:

$$\begin{aligned}
 \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{x^2 - y^2} &= \lim_{x \rightarrow 0} \frac{x^2 + (mx)^2}{x^2 - (mx)^2} = \lim_{x \rightarrow 0} \frac{1 + m^2}{1 - m^2} \\
 &= \frac{1 + m^2}{1 - m^2}
 \end{aligned}$$

So, the limit depends on m . Thus, the limit does not exist.

Example:

Discuss the existence of the $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + 3y^4}$.

Solution

Using the repeated limits:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + 3y^4} = \lim_{x \rightarrow 0} \left[\lim_{y \rightarrow 0} \left(\frac{x^2 y^2}{x^4 + 3y^4} \right) \right] = \lim_{x \rightarrow 0} \left[\frac{0}{x^4} \right] = \lim_{x \rightarrow 0} [0] = 0$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + 3y^4} = \lim_{y \rightarrow 0} \left[\lim_{x \rightarrow 0} \left(\frac{x^2 y^2}{x^4 + 3y^4} \right) \right] = \lim_{y \rightarrow 0} \left[\frac{0}{3y^4} \right] = \lim_{y \rightarrow 0} [0] = 0$$

While by using the path method:

Let $y = mx$:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + 3y^4} = \lim_{x \rightarrow 0} \frac{x^2 (mx)^2}{x^4 + 3(mx)^4} = \lim_{x \rightarrow 0} \frac{m^2}{1 + 3m^4} = \frac{m^2}{1 + 3m^4}.$$

So, the limit depends on m . Thus, the limit does not exist.